

# ELI — Coding Agent

ELI v2.2.6

August 2026

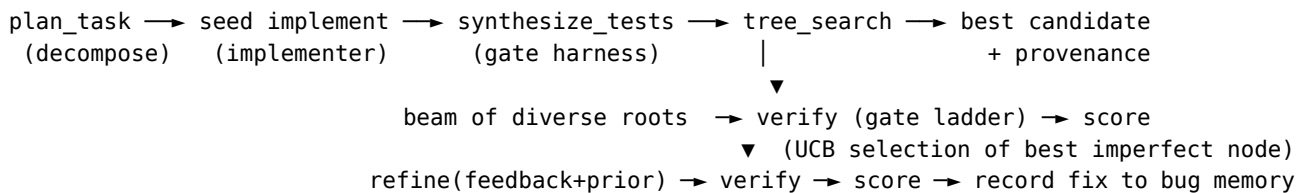
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## ELI Coding Agent (eli/coding/)

A self-contained, **additive** subsystem that lifts ELI's code generation, analysis, and repair to frontier-grade. It touches nothing that already works: it lives in its own package and is reached through a new `CODE_SOLVE` action. The LLM is an *injected* generate callable (default = the local inference broker), so the whole pipeline is model-agnostic and unit-testable without a model.

### The loop



## Components

| File       | Capability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sandbox.py | <b>Mandatory execution-feedback substrate.</b><br>Runs code in a bounded, isolated subprocess (temp cwd, scrubbed env, MPLBACKEND=Agg, wall-clock timeout, POSIX RLIMIT_CPU; <b>no RLIMIT_AS</b> — it breaks numpy). Multi-language via a runner map (Python deepest). Crash detection is precise: only a genuine Python traceback counts; timeouts / signal kills / missing optional deps are tolerated so legitimate long/heavy/optional code isn't falsely failed. <code>smoke_import</code> verifies a patched module still loads. |

| File            | Capability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bug_memory.py   | <b>Semantic bug classification + long-term memory.</b> <code>classify_bug</code> maps a failure to a <code>BugClass</code> (null-handling, type, index/off-by-one, key-missing, name-undefined, import, zero-division, value, assertion, recursion, infinite-loop, logic-inversion, state-corruption, concurrency, resource-leak, io) with a stable, fuzzy-matchable <i>signature</i> (exception type + last frame + normalized message). <code>BugMemory</code> is a local SQLite store of (signature → class → fix) with exact-then-fuzzy (diffib) recall, so the agent learns from past fixes. |
| verification.py | <b>Explicit verification gating + scoring + test synthesis.</b> <code>verify_candidate</code> runs the ladder syntax → execution → tests, stopping at the first failure and attaching a <code>BugDiagnosis</code> + repair feedback. <code>score_candidate</code> aggregates a 0..1 score (tests dominate when present, else clean execution). <code>synthesize_tests</code> produces an executable harness (LLM-driven; deterministic smoke-harness fallback) that prints ELI_TESTS: <passed>/<total>.                                                                                           |
| planner.py      | <b>Structured decomposition — planner/implementer separation.</b> <code>plan_task</code> → an approach + ordered steps (JSON, with a single-step fallback). <code>implement</code> writes/refines code against the plan; in refine mode it does <b>patch-based incremental refinement</b> (fix the specific feedback, keep prior code).                                                                                                                                                                                                                                                           |
| search.py       | <b>Tree search over solutions (not single-shot).</b> A diverse beam of roots (temperature ladder = exploration), then iterative expansion choosing the node to refine by <b>UCB1</b> ( $\text{score} + c \cdot \sqrt{(\ln N / n_i)}$ ) — exploitation vs exploration. Refinements pull matching prior fixes from bug memory into the feedback; successful repairs are written back as (bug → unified-diff fix) pairs. Stops on <code>target_score</code> or budget.                                                                                                                               |
| agent.py        | <b>CodeAgent / solve()</b> — composes all of the above; default broker-backed generator; returns best code + full provenance (plan, search trace, score, bug class).                                                                                                                                                                                                                                                                                                                                                                                                                              |

### Capability checklist (as requested)

- structured decomposition (planner/implementer separation) — `planner.py` [done]
- execution feedback loop (mandatory) — `sandbox.py` + `verify_candidate` gate 2 [done]
- tree search over solutions (not single shot) — `search.py` beam + UCB [done]
- explicit verification gating — `verification.py` gate ladder [done]
- long-term memory of bugs and fixes — `bug_memory.py` [done]
- automated test synthesis — `verification.synthesize_tests` [done]
- patch-based incremental refinement — `planner.implement` refine mode + search expansion [done]
- semantic bug classification (null handling, logic inversion, state corruption, ...) — `bug_memory.classify_bug`

[done]

- scoring / “U-style” exploration / tree-of-code-solutions — score\_candidate + UCB beam tree [done]
- multiple languages — sandbox runner map (Python deep; bash/js/ts/ruby/go/lua structural) [done]

## Wiring & control

- New executor action **CODE\_SOLVE** (args.description, optional args.language) saves the solution to artifacts/scripts/ and returns code + provenance. It is **additive** — GENERATE\_SCRIPT and the self-patch engine are untouched.
- Env knobs: ELI\_CODING\_SANDBOX (default on; 0 disables execution), ELI\_CODING\_RUN\_TIMEOUT (20s), ELI\_CODING\_BEAM (3), ELI\_CODING\_MAX\_ITERS (6).
- Bug memory DB: <db\_dir>/coding\_memory.sqlite3.

## Honest scope (what’s verified vs model-dependent)

- **Verified deterministically** (tests in tests/test\_coding\_agent.py, no model): the sandbox crash/tolerate logic, the bug classifier + memory recall, the gate ladder + scoring, and the full plan→search→repair→record loop driven by a stub generator (buggy seed → UCB refinement → correct, fix recorded).
- **Model-dependent quality:** the *content* of plans, implementations, and synthesized tests scales with the local model. The machinery is frontier-shaped regardless of model.
- **Depth:** execution-level verification is Python-first; other languages run if their interpreter is installed (structural support).
- **Subtask DAG (added):** decomposable tasks are solved as a dependency graph of subtasks on the shared eli/core/dag engine (plan\_graph.solve\_dag), composed into one module — see dag.md. Gated by ELI\_CODING\_DAG (default on); cohesive tasks fall back to single-shot.
- **Chat routing (added):** CODE\_SOLVE is now reachable from chat for explicit intents (“solve this coding problem”, “implement X and test it”, “use the coding agent”, “verified solution”). Generic “write a python function” still routes to the lighter GENERATE\_SCRIPT (an earlier router path claims it — precedence is tunable but the router was intentionally left unreordered).
- **Not yet:** composition is single-module (ordered concat + import-dedupe), not true multi-file projects; test synthesis verifies against synthesized tests, not a user-supplied spec suite.